Check for updates

Letter

Targeted mutagenesis of *HOX-A2* and *HOX-D2* enhances floret fertility and grain number in common wheat

Wheat (*Triticum aestivum* L.) was first domesticated in the Fertile Crescent *c.* 9600 BC (Cavalet-Giorsa *et al.*, 2024). As one of the most widely consumed cereals today, it serves as an important carbohydrates source for the growing global population, which is projected to reach 9.7 billion by 2050 (Hamad & Tayel, 2025). Grain yield is mainly determined by three major components: the number of spikes per unit land area, the number of grains per spike, and the thousand-grain weight. The grain number per spike, in turn, depends on the number of spikelets per spike and the number of grains per spikelet (Zhang *et al.*, 2022). Wheat spikelets are indeterminate and can develop up to 12 floret primordia along the elongated rachilla (Sibony & Pinthus, 1988; Guo & Schnurbusch, 2015). Therefore, increasing the spike fertility index, defined as the number of grains per unit spike dry weight at anthesis (Fischer, 1983; Ferrante *et al.*, 2012), represents a promising approach to improve yield potential.

Floret fertility is genetically controlled within cultivars (Guo *et al.*, 2017). Loss of *Vrs1* (*six-rowed spike 1*) function promotes the development of rudimentary lateral spikelets into fully fertile ones, converting two-rowed barley into six-rowed barley (Komatsuda *et al.*, 2007). Recent studies show that the ortholog of *Vrs1* in tetraploid durum wheat (*Triticum turgidum* ssp. *durum*), nomenclature as *Grain Number Increase 1* (*GNI-A1*), is an important contributor to floret fertility; its mutation increases floret fertility and grain number per spikelet (Sakuma *et al.*, 2019). Wheat *HOMEOBOX2* (*TaHOX2*) is a paralog of *GNI-A1*, and each gene has three homoeoalleles (*TaHOX2*: *TraesFLD2A01G218200*, *TraesFLD2B01G242000*, *TraesFLD2D01G228100*, and *GNI1*: *TraesFLD2A01G428900*, *TraesFLD2B01G437500*, *TraesFLD2D01G423200*) in common wheat (Gallagher *et al.*, 2023). These six homoeoalleles share over 82.21% and 75.73% sequence identity at the nucleotide and protein levels, respectively (Supporting Information Figs S1, S2; Table S1). The biological function of *TaHOX2* in regulating floret fertility remains unexplored.

Clustered regularly interspaced short palindromic repeats/CRISPR-associated 9 (CRISPR/Cas9) technology has been extensively utilized in cereal crop improvement, including in rice (*Oryza sativa* L.), barley (*Hordeum vulgare* L.), maize (*Zea mays* L.), and wheat (Wang & Doudna, 2023). In this study, we applied this system in wheat to clarify the biological function of *TaHOX2* homoeoalleles in regulating floret fertility and grain number. *TaHOX2* encodes a homeodomain leucine zipper class I (HD-Zip I) transcription factor, and its three homoeoalleles share > 96% protein sequence identity (Fig. S2). The gene structure consists of three exons interspaced by two introns (Fig. 1b). Two single-guide RNA target sites were selected within the second exon's consensus region across all homoeoalleles for editing (Fig. 1b).

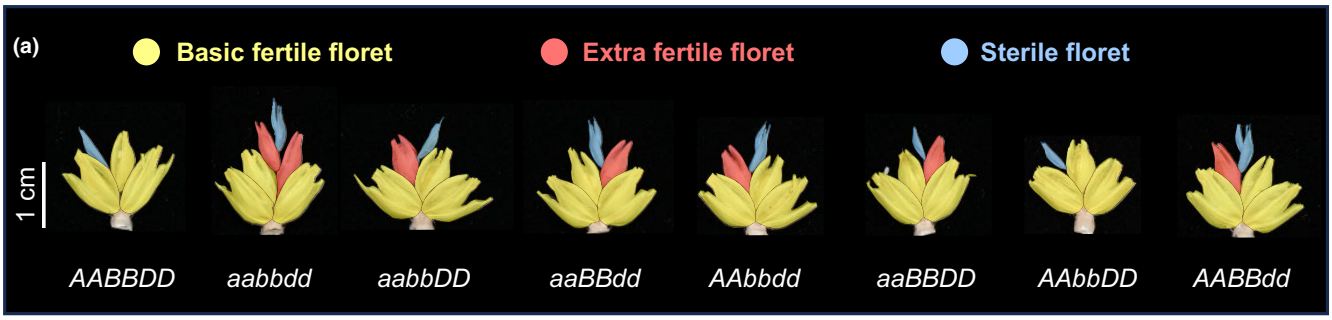
Using the wheat cultivar 'Fielder' for transformation, we obtained 94 T0 regenerants, among which 39 carried mutations at one or more of the preselected target sites across the homoeoalleles (Table S2). Subsequent screening of the T1 and T2 generations yielded homozygous mutants for all *TaHOX2* homoeoalleles, including single (*aaBBDD*, *AAbbDD*, and *AABBdd*), double (*aabbDD*, *aaBBdd*, and *AAbbdd*), and triple (*aabbdd*) mutant genotypes (Figs 1b, S3; Table S3). Notably, T-DNA-free mutants identified in the T1 generation were utilized for downstream analyses to eliminate potential developmental side effects caused by T-DNA integration or unintended CRISPR/Cas9-induced mutations (Fig. S4). In addition, we checked potential off-target effects and revealed no mutations in homoeoalleles of *GNI1*, the paralog of *TaHOX2* (Fig. S4).

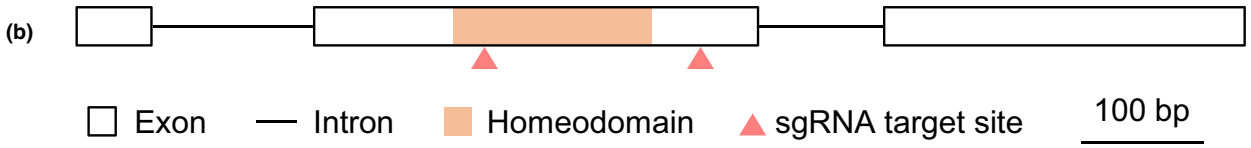
Subsequently, we assessed 12 agronomic traits under two growing conditions: (1) a controlled environment room (CER) maintained at a 16 h : 8 h, 22°C : 17°C, light : dark photoperiod, 300 μmol m^{-2} s^{-1} light intensity, and 60% relative humidity. Plants were grown in individual square pots (14 × 14 × 12 cm) filled with PINDSTRUP substrate and vermiculite mixed at a 30 : 7 (v/v) ratio; and (2) a field environment in Weifang, China (36°34′38″N, 119°24′40″E). Nine genotypes were phenotyped, including seven mutants, the unedited wild-type (WT) derived from the edited lines, and the cultivar Fielder. For the CER trials, each genotype was represented by 16 replicates arranged in a completely randomized design (Fig. S5; Table S4). In the field experiment, 40 grains were evenly sown per two-meter row, with three replicates per genotype. Distance between each rows were spaced 25 cm apart. As Fielder is a spring cultivar, the field experiment was conducted from mid-March to the end of June 2025. The set of measured traits was consistent across both environments (Figs 1c, S5, S6; Tables S5–S8).

The single, double, and triple mutants used for phenotyping were all derived from T0 plant SpCas9-2_46. Mutations in the *HOX-A2* and *HOX-B2* homoeoalleles disrupted the homeodomain and thus completely abolished their respective protein function. By contrast, *HOX-D2* carried a lesion with a 1-bp deletion occurring downstream of the homeodomain. To confirm that *HOX-D2* in line SpCas9-2_46_12_8 (designated *aabbdd-1*) is a complete lost-of-function allele, we included two additional independent triple mutants, SpCas9-2_46_37_5 and SpCas9-2_66_14_9, named *aabbdd-2* and *aabbdd-3*, respectively. Both individuals are *hox2* null mutants as the homeodomain is disrupted in all three

© 2026 The Author(s).
New Phytologist © 2026 New Phytologist Foundation.

New Phytologist





ABD

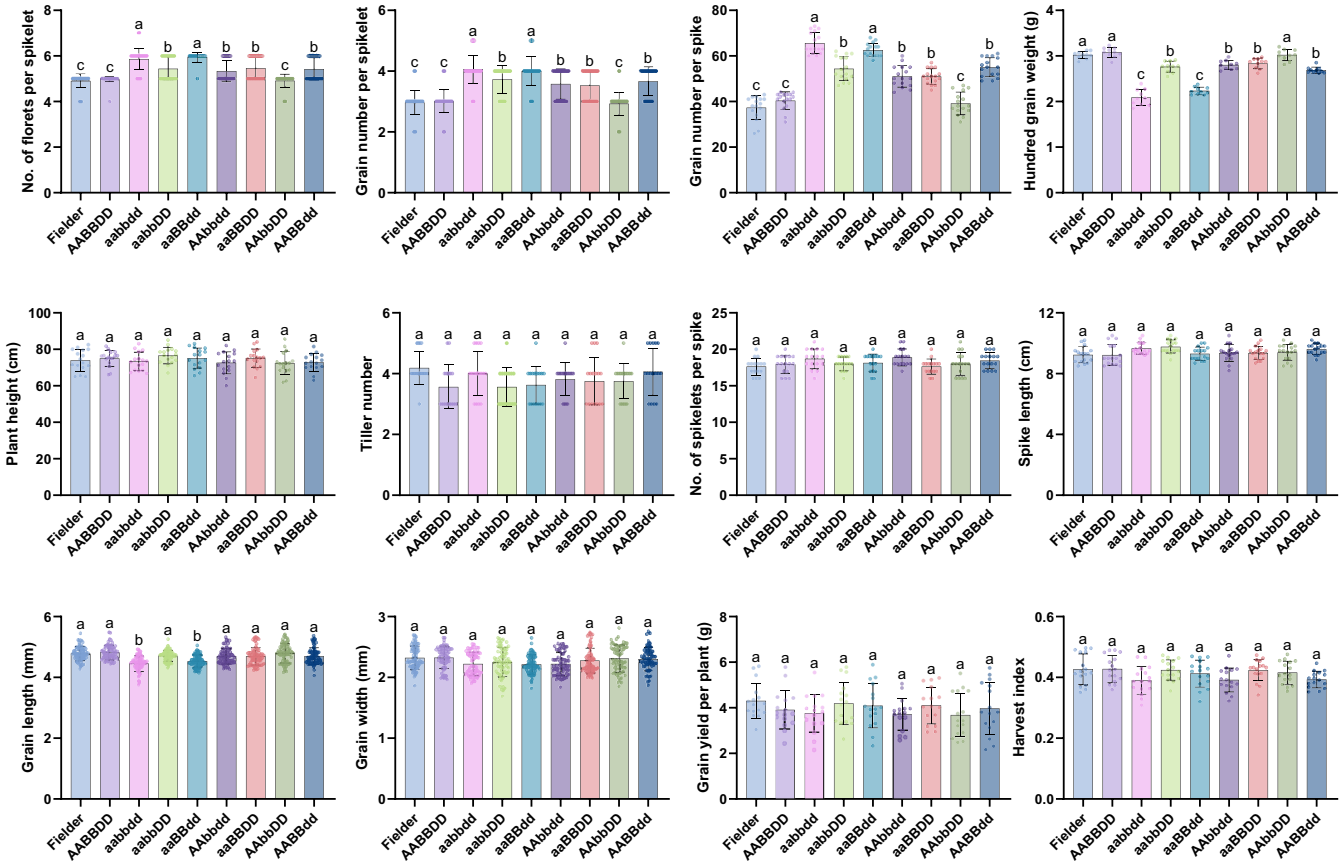
5′-CGGCTGACCGACGAGCAGGTGCACGGTCTGNNNNNNNNNNNNNGCTCAAGCACGCCCACGACGCGGCC-3′
3′-GCCGACTGGCTGCTCGTCCACGTGCCAGACNNNNNNNNNNNNNCGAGTTCGTGCGGGTGCTGCGCCGG-5′

sgRNA2 PAM2
PAM1 sgRNA1

aabbdd mutant

		Mutation	WT allele
A	5′-CGGCTGACCGACG-GCAGGTGCACGGTCTGNNNNNNNNNNNNNGCTCAAGCACGCCC---ACGCGGCC-3′	–1 bp & –3 bp	–
B	5′-CGGCTGACCGACG-GCAGGTGCACGGTCTGNNNNNNNNNNNNNGCTCAAGCACGCCCAC-ACGCGGCC-3′	–1 bp & –1 bp	–
D	5′-CGGCTGACCGACGAGCAGGTGCACGGTCTGNNNNNNNNNNNNNGCTCAAGCACGCCCAC-ACGCGGCC-3′	–1 bp	–

(c)



© 2026 The Author(s).
New Phytologist © 2026 New Phytologist Foundation.

14698137, 0, Downloaded from https://nph.onlinelibrary.wiley.com/doi/10.1111/nph.71233 by Tianjin Normal University, Wiley Online Library on [01/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

14698137, 0, Downloaded from https://nph.onlinelibrary.wiley.com/doi/10.1111/nph.71233 by Tianjin Normal University, Wiley Online Library on [01/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Fig. 1 Site-directed mutagenesis of *TaHOX2* homoeologs enhances floret fertility and grain number in wheat. (a) Spikelet phenotype of wild-type (WT) and *TaHOX2* single, double, and triple mutants. Using the WT as a reference, the three basal fertile florets are designated as basic fertile florets. The basic fertile florets, extra fertile florets in the mutants, and sterile florets are highlighted in pseudocolor (yellow, pink, and blue, respectively). (b) *TaHOX2* gene structure with single-guide RNA target sites and triple mutant allele modifications. PAM: protospacer adjacent motif. (c) Phenotypic characterization of 12 agronomic traits under the controlled environment room (CER) conditions. Individual measurements are shown as dots. Hundred-grain weight (HGW) was measured for CER-grown plants due to limited grain availability. AABBDD represents WT for all threeTaHOX2 homoeoalleles, while the lowercase letters represent mutant of the respective homoeoalleles. Statistical significance was determined by one-way ANOVA with Tukey's test ($P = 0.05$). Error bars represent SD. Number of florets per spikelet and grain number per spikelet ($n = 80$); grain number per spike, plant height, tiller number, spikelet number per spike, spike length, grain yield per plant, and harvest index ($n = 16$); HGW ($n = 10$); grain length and grain width ($n = 60$).

homoeoalleles (Fig. S7). Assessment of floret number per spikelet and grain number per spikelet revealed no difference among the three triple mutants (Fig. S7C). Therefore, *HOX-D2* represents a dysfunctional allele in line *aabbdd-1*.

The loss of *HOX-A2* function (genotype *aaBBDD*) enhanced floret fertility in hexaploid common wheat (Figs 1, S3, S5; Tables S5–S8). Similarly, loss of *HOX-D2* function (*AABBdd*) increased the number of florets per spikelet to a level comparable to that of the *aaBBDD* mutant. Conversely, the *HOX-B2* homoeoallele (*AAbbDD*) had negligible effects on floret fertility (Figs 1, S3, S6; Tables S5–S8). The double mutants *aabbDD* and *AAbbdd* maintained floret counts similar to their corresponding single mutants. Intriguingly, the *aaBBdd* double mutant exhibited more florets per spikelet than any other single or double mutant, while the tripe mutant *aabbdd* achieved the highest floret count, although it was not significantly different from *aaBBdd* (Figs 1, S3, S6; Tables S5–S8).

Increased floret numbers improved grain number in the mutants. At the whole-spike level, both florets and grains per spike increased significantly, exhibiting significance patterns mirroring those of florets per spikelet (Figs 1, S3, S6; Tables S5–S8). Spike length and the number of spikelets per spike remained unchanged, suggesting that the increase in florets and grains per spikelet was primarily driven by the spike-level increases (Figs 1, S3, S6; Tables S5–S8). In addition, *TaHOX2* mutations had no effect on plant height or tiller number.

Grains from the fourth or higher order florets were notably smaller in the mutants. Accordingly, both hundred-grain weight and thousand-grain weight, measured under CER and field conditions, respectively, decreased in mutants containing loss-of-function alleles in *HOX-A2* or *HOX-D2* (Figs 1, S3, S6; Tables S5–S8). To mitigate confounding effects from under-developed grains, length and width were quantified using well-developed grains from the two basal florets. This measurement revealed a restriction in longitudinal growth, yielding smaller grains with reduced grain length in the *aabbdd* and *aaBBdd* mutants (Figs 1, S3, S6, S8; Tables S5–S8). Ultimately, however, grain yield per plant and the harvest index exhibited no significant differences between the WT and the mutants (Figs 1, S3, S6; Tables S5–S8).

In conclusion, this study elucidates the functional roles of *TaHOX2* homoeoalleles in determining floret fertility and grain number in common wheat. Our work demonstrates that *HOX-A2* and *HOX-D2* negatively modulate floret development and fertility, and their loss of function leads to an increase in floret and grain number per spikelet. By contrast, *HOX-B2* appears to be functionally dispensable in this process, which may be attributed to its lower expression level related to other homoeoalleles (Fig. S9). Similar to *HOX-B2*, *GNI-B1* exhibits minimal expression in hexaploid wheat (Sakuma *et al.*, 2019). In barley, HvHOX1 and HvHOX2 proteins form heterodimers (Thirulogachandar *et al.*, 2024). It is plausible that TaHOX2 interacts with TaHOX1/GNI1 to regulate floret fertility and grain number. Collectively, the genetic dissection of *TaHOX2* advances our understanding of inflorescence development and provides critical insight for breeding higher-yielding varieties. Nevertheless, optimizing trade-offs, such as the increase in grain number at the expense of grain size in *TaHOX2* mutants, will be essential in future molecular design breeding. Further integration of strategies aimed at enhancing photosynthetic efficiency may help unlock additional yield potential in such lines.

Materials and Methods

Plasmid construction and wheat transformation

Protospacer sequences (PS) targeting all three *TaHOX2* homoeoalleles were selected by help of the online tool CRISPR-GE. Specificity of PS to *TaHOX2* was confirmed by BLAST against the wheat reference genome to rule out potential off-target effects. The PS was introduced into GRF4-GIF1 vector (Pan *et al.*, 2025) through In-Fusion cloning, followed by transformation into *Agrobacterium* strain EHA105. Wheat variety 'Fielder' was used as recipient material for *Agrobacterium*-mediated transformation.

Genotyping of CRISPR/Cas9-induced mutations

We designed *TaHOX2* A-/B-/D-specific primers using the online tool GSP (Genome-specific Primers) (Wang *et al.*, 2016). Pre-selected target sites were PCR-amplified; subsequently, screening of CRISPR/Cas9-induced edits was confirmed by Sanger sequencing. Decoding of mutations in T0 to T2 plants was achieved by the help of the GENEIOUS software.

Acknowledgements

The authors gratefully acknowledge Shijuan Zhang for her technique support; the Biotechnology platform at PKU-IAAS for wheat transformation. The work is supported by the National Key R&D Program of China (grant 2024YFF1000301), the Taishan Scholars Program (grant tsqn202312306), and the Shandong Provincial Natural Science Fund for Excellent Young Scientists Fund Program (Overseas) (grant 2025HWYQ-071) to ML, and

© 2026 The Author(s).
New Phytologist © 2026 New Phytologist Foundation.

the Shandong Provincial Natural Science Foundation (grants SYS202206, ZR2022ZD22, and ZR2021ZD30).

Competing interests

None declared.

Author contributions

YJ conceived the study. ML coordinated the project. HZ and GG performed the experiments. ML and HZ analyzed the data. ML wrote the initial draft. ML and YJ revised the manuscript. All authors read and approved the manuscript.

ORCID

Guofeng Gao https://orcid.org/0009-0002-6924-7506
Yuling Jiao https://orcid.org/0000-0002-1189-1676
Mingjiu Li https://orcid.org/0000-0003-3299-5287
Haiyan Zhang https://orcid.org/0009-0009-0145-5123

Data availability

The data that support the findings of this study are available in the Supporting Informationof this article (Figs S1–S9; Tables S1–S8).

Mingjiu Li (李明久)[1]*, Haiyan Zhang (张海艳)[1], Guofeng Gao (高国峰)[1] and Yuling Jiao (焦雨铃)[1,2,3]*

[1]State Key Laboratory of Wheat Improvement, Peking University Institute of Advanced Agricultural Sciences, Shandong Laboratory of Advanced Agricultural Sciences in Weifang, Weifang, Shandong, 261325, China;
[2]State Key Laboratory for Gene Function and Modulation Research, School of Life Sciences, Peking University, Beijing, 100871, China;
[3]Peking-Tsinghua Center for Life Sciences, Center for Quantitative Biology, Academy for Advanced Interdisciplinary Studies, Peking University, Beijing, 100871, China
(*Authors for correspondence: email mingjiu.li@pku-iaas.edu.cn (ML); yuling.jiao@pku.edu.cn (YJ))

References

Cavalet-Giorsa E, González-Muñoz A, Athiyannan N, Holden S, Salhi A, Gardener C, Quiroz-Chávez J, Rustamova SM, Elkot AF, Patpour M *et al.* 2024. Origin and evolution of the bread wheat D genome. *Nature* **633**: 848–855.

Ferrante A, Savin R, Slafer G. 2012. Differences in yield physiology between modern, well adapted durum wheat cultivars grown under contrasting conditions. *Field Crops Research* **136**: 52–64.

Fischer R. 1983. Wheat. In: Smith WH, Banta SJ, eds. *Symposium on potential productivity of field crops under different environments.* Los Baños, Philippines: International Rice Research Institute, 129–153.

Gallagher JP, Man J, Chiaramida A, Rozza IK, Patterson EL, Powell MM, Schrager-Lavelle A, Multani DS, Meeley RB, Bartlett ME. 2023. *GRASSY TILLERS1* (*GT1*) and *SIX-ROWED SPIKE1* (*VRS1*) homologs share conserved roles in growth repression. *Proceedings of the National Academy of Sciences, USA* **120**: e2311961120.

Guo Z, Chen D, Alqudah AM, Roder MS, Ganal MW, Schnurbusch T. 2017. Genome-wide association analyses of 54 traits identified multiple loci for the determination of floret fertility in wheat. *New Phytologist* **214**: 257–270.

Guo Z, Schnurbusch T. 2015. Variation of floret fertility in hexaploid wheat revealed by tiller removal. *Journal of Experimental Botany* **66**: 5945–5958.

Hamad A, Tayel A. 2025. Food 2050 concept: trends that shape the future of food. *Journal of Future Foods* **6**: 1053–1066.

Komatsuda T, Pourkheirandish M, He C, Azhaguvel P, Kanamori H, Perovic D, Stein N, Graner A, Wicker T, Tagiri A *et al.* 2007. Six-rowed barley originated from a mutation in a homeodomain-leucine zipper I-class homeobox gene. *Proceedings of the National Academy of Sciences, USA* **104**: 1424–1429.

Pan W, Gao C, Niu D, Cheng J, Zhang J, Yan X, Long Q, Zhu Y, Sun W, Xie Q *et al.* 2025. Efficient gene disruption in polyploid genome by Cas9-Trex2 fusion protein. *Journal of Integrative Plant Biology* **67**: 7–10.

Sakuma S, Golan G, Guo Z, Ogawa T, Tagiri A, Sugimoto K, Bernhardt N, Brassac J, Mascher M, Hensel G *et al.* 2019. Unleashing floret fertility in wheat through the mutation of a homeobox gene. *Proceedings of the National Academy of Sciences, USA* **116**: 5182–5187.

Sibony M, Pinthus MJ. 1988. Floret initiation and development in spring wheat (*Triticum aestivum* L.). *Annals of Botany* **61**: 473–479.

Thirulogachandar V, Govind G, Hensel G, Kale SM, Kuhlmann M, Eschen-Lippold L, Rutten T, Koppolu R, Rajaraman J, Palakolanu SR *et al.* 2024. *HOMEOBOX2*, the paralog of *SIX-ROWED SPIKE1*/*HOMEOBOX1*, is dispensable for barley spikelet development. *Journal of Experimental Botany* **75**: 2900–2916.

Wang JY, Doudna JA. 2023. CRISPR technology: a decade of genome editing is only the beginning. *Science* **379**: eadd8643.

Wang Y, Tiwari VK, Rawat N, Gill BS, Huo N, You FM, Coleman-Derr D, Gu YQ. 2016. GSP: a web-based platform for designing genome-specific primers in polyploids. *Bioinformatics* **32**: 2382–2383.

Zhang X, Jia H, Li T, Wu J, Nagarajan R, Lei L, Powers C, Kan CC, Hua W, Liu Z *et al.* 2022. *TaCol-B5* modifies spike architecture and enhances grain yield in wheat. *Science* **376**: 180–183.

Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Nucleotide sequence alignment of *TaHOX2* and *GNI1* homoeologs.

Fig. S2 Protein sequence alignment of TaHOX2 and GNI1 homoeologs.

Fig. S3 Mutation detection in *TaHOX2* alleles across T0 to T2 generations.

Fig. S4 Analysis of sgRNA specificity and T-DNA presence.

Fig. S5 Images of CER and field trials.

Fig. S6 Phenotypic characterization of 12 agronomic traits under field conditions.

Fig. S7 Comparison of floret number per spikelet and grain number per spikelet in three independent *hox2* triple mutants (*aabbdd*).

Fig. S8 Measurement of grain length and grain width.

© 2026 The Author(s).
New Phytologist © 2026 New Phytologist Foundation.

14698137, 0, Downloaded from https://nph.onlinelibrary.wiley.com/doi/10.1111/nph.71233 by Tianjin Normal University, Wiley Online Library on [01/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Fig. S9 Expression profiles of *TaHOX2* homoeologs across various tissues and developmental stages.

Table S1 Nucleotide and amino acid sequence similarity between *TaHOX2* and *GNI1* homoeologs.

Table S2 Summary of genome editing events in T0 transgenic plants.

Table S3 List of primers used in this study.

Table S4 Experimental design for the phenotyping under CER conditions.

Table S5 Raw phenotypic data from the CER experiment.

Table S6 Statistical analysis of phenotypic data from the CER experiment.

Table S7 Raw phenotypic data from the field experiment.

Table S8 Statistical analysis of phenotypic data from the field experiment.

Please note: Wiley is not responsible for the content or functionality of any Supporting Information supplied by the authors. Any queries (other than missing material) should be directed to the *New Phytologist* Central Office.

Key words: floret fertility, genome editing, grain number, *TaHOX2*, wheat.

Received, 2 December 2025; accepted, 15 April 2026.

Disclaimer: The New Phytologist Foundation remains neutral with regard to jurisdictional claims in maps and in any institutional affiliations.

© 2026 The Author(s).
New Phytologist © 2026 New Phytologist Foundation.

14698137, 0, Downloaded from https://nph.onlinelibrary.wiley.com/doi/10.1111/nph.71233 by Tianjin Normal University, Wiley Online Library on [01/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License